LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Open loop and closed loop system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Open loop and closed loop system.
5. Specific Objectives	20.1.1.1 Trainee will be able to classify control systems, feedback and its effect.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Draw block diagrams of open and closed loop systems; Safety: Ensure diagrams are neat and labeled; Supervision: Instructor verifies accuracy of feedback paths.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft control system?
(b) Can you name one example of a mechanical linkage in an aircraft?
(c) Why is system classification important for maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of open loop system • Characteristics of open loop control • Examples in aircraft systems • Limitations of open loop systems	L & Ds	PPT, IB	15 min
(b)	• Definition of closed loop system • Role of feedback in control • Comparison of open and closed loop • Advantages of closed loop systems	L & Ds	PPT, PDF	20 min
(c)	• Effect of feedback on stability • Effect of feedback on accuracy • Real-world aircraft applications • Summary of classification criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) How does feedback improve system accuracy?
(b) What is the main disadvantage of an open loop system?
(c) Give an example of a closed loop system in an aircraft.

12. Home Assignment and Follow Up	Prepare a brief note comparing the stability characteristics of open and closed loop systems to be discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Feedback, types and its effect on control system.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min